

Problem Set 1

Matrices, finite differences, elimination, and point loads

18.085 Computational Science and Engineering I • Fall 2026 • G. G. Rozenman

Due Thursday, October 1, 2026, at 11:59 pm ET

Covers Lectures 1–5 (Strang §1.1–1.4). Five problems, 120 points.

Collaboration. You are encouraged to work together, but all final write-ups must be your own. List your discussion partners and any outside resources you consulted, including AI tools. Using an AI system in place of your own reasoning without saying so violates the collaboration policy; using one as a study aid and saying so does not.

Computing. Problem 2(d)–(e) is computational. Use Python or MATLAB, your choice, starting from `pset1_starter.py` or `pset1_starter.m` on Canvas. Attach your code, your plot, and your table of slopes to the PDF.

1. Multiple views of matrix multiplication (20 pts)

★**

From: L1, L4 · Strang 1.1

Consider the matrices

$$\mathbf{A} = \begin{bmatrix} 3 & -2 \\ 1 & 4 \\ 1 & 1 \end{bmatrix} \quad (3 \times 2), \quad \mathbf{B} = \begin{bmatrix} 2 & 5 & -1 \\ -1 & 3 & 2 \end{bmatrix} \quad (2 \times 3).$$

Using the 3×3 product $\mathbf{AB}$ as a concrete example, we look at two ways to view matrix multiplication that are useful but may be less familiar than “row times column.”

- (a) **Linear combinations of the columns.** When $\mathbf{A}$ multiplies a column vector $\mathbf{v}$, the result is a combination of the columns $\mathbf{a}_1, \mathbf{a}_2$ of $\mathbf{A}$ with the entries of $\mathbf{v}$ as coefficients: $\mathbf{Av} = v_1\mathbf{a}_1 + v_2\mathbf{a}_2$. Compute $\mathbf{AB}$ from this point of view,

$$\mathbf{AB} = \mathbf{A} \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \mathbf{b}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{Ab}_1 & \mathbf{Ab}_2 & \mathbf{Ab}_3 \end{bmatrix},$$

writing each column as an explicit combination of $\mathbf{a}_1$ and $\mathbf{a}_2$.

- (b) **Sums of outer products.** Let $\mathbf{b}'_1{}^\top$ and $\mathbf{b}'_2{}^\top$ be the two *rows* of $\mathbf{B}$. Confirm that

$$\mathbf{AB} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 \end{bmatrix} \begin{bmatrix} \mathbf{b}'_1{}^\top \\ \mathbf{b}'_2{}^\top \end{bmatrix} = \mathbf{a}_1\mathbf{b}'_1{}^\top + \mathbf{a}_2\mathbf{b}'_2{}^\top$$

by computing each term. Each term is a 3×3 matrix formed from a column times a row, an *outer product*. Without computing a determinant, use (a) or (b) to explain why the 3×3 matrix $\mathbf{AB}$ cannot be invertible.

- (c) **The column view in reverse.** Let $\mathbf{W} = \begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$ and $\mathbf{d} = \begin{bmatrix} 2 \\ 11 \end{bmatrix}$. Since $\mathbf{W}\mathbf{x} = x_1\mathbf{w}_1 + x_2\mathbf{w}_2$, solving $\mathbf{W}\mathbf{x} = \mathbf{d}$ means finding the coordinates of $\mathbf{d}$ in the basis $\{\mathbf{w}_1, \mathbf{w}_2\}$ (Lecture 4). The columns of $\mathbf{W}$ are orthogonal but not of unit length. Check that $\mathbf{w}_1 \perp \mathbf{w}_2$, find x_1 and x_2 by

projection, without elimination, and confirm that $\mathbf{W}\mathbf{x} = \mathbf{d}$. Then apply the same projection formula to the columns of $\mathbf{A}$ and the vector $\mathbf{z} = \mathbf{A} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$. Do you recover the coefficients 1, 1? Explain.

2. Finite difference convergence rates (40 pts)

★★★

From: L1–L2 · Strang 1.2

Assume $u(x)$ is smooth (as many derivatives as needed). Write

$$D_h u(x) = \frac{u(x+h) - u(x-h)}{2h}$$

for the centred first difference with step h .

(a) **One more term.** Using Taylor series, show that

$$D_h u(x) = u'(x) + \frac{h^2}{6} u'''(x) + \frac{h^4}{120} u^{(5)}(x) + \mathcal{O}(h^6).$$

Why do only even powers of h appear? Without any further expansion, write down the corresponding result for $D_{2h} u(x) = \frac{u(x+2h) - u(x-2h)}{4h}$. Both are $\mathcal{O}(h^2)$; how do their leading errors compare?

(b) **A one-sided second difference.** Consider

$$u''(x) \approx \frac{u(x) - 2u(x-h) + u(x-2h)}{h^2}.$$

Find the order of accuracy of this approximation and its leading error term, and compare with the three-point centred scheme $(u(x+h) - 2u(x) + u(x-h))/h^2$. Lecture 1 built the centred stencil as the product $\mathbf{AB}$ of the backward difference matrix $\mathbf{A}$ and the forward difference matrix $\mathbf{B}$. Which of $\mathbf{AB}$, $\mathbf{BA}$, $\mathbf{A}^2$, $\mathbf{B}^2$ has the stencil above in its interior rows?

(c) **An uneven stencil.** Derive a formula

$$u'(x) \approx \frac{a u(x-h) + b u(x) + c u(x+2h)}{h}$$

whose error is $\mathcal{O}(h^2)$. (A stencil like this appears next to a boundary, or where the grid spacing changes.) Write the conditions on a, b, c as a 3×3 linear system and solve it. What is the leading error term?

(d) **Computational.** Check (a)–(c) on $u(x) = \sin 2x$ at $x = 0.5$, for $h = 2^{-1}, 2^{-2}, \dots, 2^{-12}$. Make one log–log plot of |error| against h with a curve for each of: D_h , D_{2h} , the two second-derivative formulas in (b), and your formula from (c). Fit the slope of each curve by least squares on the range where it is a straight line, say which range you used, and report the slopes in a table.

(e) **Richardson extrapolation.** Find the combination $R_h u = \alpha D_h u + \beta D_{2h} u$ with $\alpha + \beta = 1$ that cancels the h^2 term, and use (a) to show that

$$R_h u(x) = u'(x) - \frac{h^4}{30} u^{(5)}(x) + \mathcal{O}(h^6).$$

Write $R_h u(x)$ as a five-point formula in $u(x \pm h)$ and $u(x \pm 2h)$, and add it to your plot. Over what range of h does its error have slope 4, and what stops it at the smallest h ?

3. LU and solving linear systems (20 pts)



From: L3 · Strang 1.3

In this problem ℓ_{ij} ($i > j$) denotes the entry of $\mathbf{L}$ in row i , column j : the multiple of row j that elimination *subtracts* from row i . (Lecture 3 recorded the multiple *added*, which is $-\ell_{ij}$; the matrices are the same.)

- (a) If $\mathbf{A} = \mathbf{LU}$, solve $\mathbf{Ax} = \mathbf{f}$ by hand without ever finding $\mathbf{A}$ itself: first solve $\mathbf{Lc} = \mathbf{f}$ (forward substitution), then $\mathbf{Ux} = \mathbf{c}$ (back substitution). Then $\mathbf{LUx} = \mathbf{Lc} = \mathbf{f}$ is the equation you wanted.

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & 2 & 1 \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 4 & 3 \\ 0 & 0 & -2 \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} 2 \\ 5 \\ -8 \end{bmatrix}.$$

- (b) **One column at a time.** Compute $\mathbf{L}_1 \mathbf{S}_1$ to verify that

$$\mathbf{L}_1 = \begin{bmatrix} 1 & & & \\ \ell_{21} & 1 & & \\ \ell_{31} & 0 & 1 & \\ \ell_{41} & 0 & 0 & 1 \end{bmatrix} \text{ is the inverse of } \mathbf{S}_1 = \begin{bmatrix} 1 & & & \\ -\ell_{21} & 1 & & \\ -\ell_{31} & 0 & 1 & \\ -\ell_{41} & 0 & 0 & 1 \end{bmatrix}.$$

(Write $\mathbf{L}_1 = \mathbf{I} + \boldsymbol{\ell} \mathbf{e}_1^\top$ with $\boldsymbol{\ell} = (0, \ell_{21}, \ell_{31}, \ell_{41})^\top$. Why is $(\boldsymbol{\ell} \mathbf{e}_1^\top)(\boldsymbol{\ell} \mathbf{e}_1^\top) = \mathbf{0}$?)

- (c) **Two columns.** Unlike (b), which involved only one column, show that, unless $\ell_{21}\ell_{32} = 0$,

$$\mathbf{L} = \begin{bmatrix} 1 & & \\ \ell_{21} & 1 & \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix} \text{ is not the inverse of } \mathbf{S} = \begin{bmatrix} 1 & & \\ -\ell_{21} & 1 & \\ -\ell_{31} & -\ell_{32} & 1 \end{bmatrix}.$$

Write $\mathbf{L} = \mathbf{L}_1 \mathbf{L}_2$ with

$$\mathbf{L}_1 = \begin{bmatrix} 1 & & \\ \ell_{21} & 1 & \\ \ell_{31} & 0 & 1 \end{bmatrix}, \quad \mathbf{L}_2 = \begin{bmatrix} 1 & & \\ 0 & 1 & \\ 0 & \ell_{32} & 1 \end{bmatrix}$$

(check this product, and compare with $\mathbf{L}_2 \mathbf{L}_1$), and compute the correct inverse $\mathbf{L}^{-1} = \mathbf{L}_2^{-1} \mathbf{L}_1^{-1}$. Notice the order. Apply your formula to the $\mathbf{L}$ of part (a), and check that $\mathbf{L}^{-1} \mathbf{f}$ reproduces the $\mathbf{c}$ you found there.

- (d) **A sparse $\mathbf{L}$ with a full inverse.** Lecture 3 factored $\mathbf{K}_4 = \mathbf{LDL}^\top$ with

$$\mathbf{L} = \begin{bmatrix} 1 & & & \\ -\frac{1}{2} & 1 & & \\ & -\frac{2}{3} & 1 & \\ & & -\frac{3}{4} & 1 \end{bmatrix}.$$

Compute $\mathbf{L}^{-1}$ and guess a formula for its entries. For the $n \times n$ version, how many nonzero entries do $\mathbf{L}$ and $\mathbf{L}^{-1}$ have? Use this to explain why one solves $\mathbf{Lc} = \mathbf{b}$ by forward substitution rather than by forming $\mathbf{L}^{-1}$ and multiplying.

4. LDL^T practice (20 pts)



From: L3 · Strang 1.3

Factor these symmetric matrices as $\mathbf{A} = \mathbf{LDL}^T$, with $\mathbf{L}$ lower triangular with ones on the diagonal and the pivots in the diagonal matrix $\mathbf{D}$:

$$\mathbf{A}_1 = \begin{bmatrix} 3 & -6 \\ -6 & 10 \end{bmatrix}, \quad \mathbf{A}_2 = \begin{bmatrix} a & b \\ b & a \end{bmatrix}, \quad \mathbf{A}_3 = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 3 & -2 \\ 0 & -2 & 5 \end{bmatrix}.$$

- (a) Factor $\mathbf{A}_1$.
- (b) Factor $\mathbf{A}_2$. What do you have to assume about a ? If $a = 0$ and $b \neq 0$, is the matrix singular? For which a, b is the second pivot zero, and what does that say about $\mathbf{A}_2$?
- (c) Factor $\mathbf{A}_3$.
- (d) **The pivots are conductances.** Show that $\mathbf{A}_3$ is the matrix of the resistor ladder in Lecture 3 with the left end open (as for $\mathbf{T}$) and the right end grounded, but with conductances 1, 2, 3 siemens from left to right instead of 1 each. Use the picture “a pivot is the conductance you have left” to explain why the pivots came out equal to the conductances. Then write down the pivots of

$$\mathbf{A}_4 = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 3 & -2 & 0 \\ 0 & -2 & 5 & -3 \\ 0 & 0 & -3 & 7 \end{bmatrix}$$

without doing any elimination.

5. Point loads and the inverse matrix (20 pts)



From: L2, L3, L5 · Strang 1.2, 1.4

Take the fixed-fixed rod

$$-u''(x) = f(x), \quad 0 < x < 1, \quad u(0) = u(1) = 0,$$

and its finite difference version $\frac{1}{h^2} \mathbf{K}_n \mathbf{u} = \mathbf{f}$ on the grid $x_i = ih$, $h = \frac{1}{n+1}$, $i = 1, \dots, n$, where u_i approximates $u(x_i)$.

- (a) Following the four steps of Lecture 5, find the Green's function $g(x, y)$: the solution of

$$-\frac{d^2 g}{dx^2} = \delta(x - y), \quad g(0, y) = g(1, y) = 0,$$

for a load at $y \in (0, 1)$. Show that $g(x, y) = x(1 - y)$ for $x \leq y$ and $g(x, y) = y(1 - x)$ for $x \geq y$. Sketch $g(x, 0.2)$ and $g(x, 0.6)$ as functions of x on the same axes, and check that $g(x, y) = g(y, x)$.

- (b) On the grid, a unit point load at node j must carry unit total load, $\sum_i f_i h = 1$, just as each rectangle in Lecture 5 had unit area. Show that this makes $\mathbf{f} = \mathbf{e}_j/h$, and that the discrete response is $\mathbf{u} = h \mathbf{K}_n^{-1} \mathbf{e}_j$, that is, h times column j of $\mathbf{K}_n^{-1}$. (The aside in Lecture 5 left out this factor of $1/h$; it matters as soon as you compare with g .) Then use superposition to write $\mathbf{u}$ for a general load vector $\mathbf{f}$ as a sum over the columns of $\mathbf{K}_n^{-1}$. Which integral from Lecture 5 is this sum approximating?

- (c) For $n = 4$, verify that

$$\mathbf{G} = \frac{1}{5} \begin{bmatrix} 4 & 3 & 2 & 1 \\ 3 & 6 & 4 & 2 \\ 2 & 4 & 6 & 3 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

satisfies $\mathbf{K}_4 \mathbf{G} = \mathbf{I}$. Its entries are $G_{ij} = \min(i, j) (n + 1 - \max(i, j)) / (n + 1)$. Show that the matrix with these entries satisfies $\mathbf{K}_n \mathbf{G} = \mathbf{I}$ for every n . (Hint: fix a column j . On each side of $i = j$ the entries G_{ij} are linear in i , just as $g(x, y)$ is linear in x on each side of $x = y$. What does a row of $\mathbf{K}_n$ do to three equally spaced values on a straight line? Then check the row $i = j$ separately.) Conclude that $h(\mathbf{K}_n^{-1})_{ij} = g(x_i, x_j)$ exactly.

- (d) Problem 2 shows that the three-point second difference has an $\mathcal{O}(h^2)$ error, yet in (c) the discrete solution has no error at all at the nodes. Explain why. Now put the load at $y = 0.25$, between the nodes $x_1 = 0.2$ and $x_2 = 0.4$ of the $n = 4$ grid. Compare two ways to represent it on the grid: the whole load at the nearest node x_1 , or the load split between x_1 and x_2 with weights $\frac{3}{4}$ and $\frac{1}{4}$ (the closer node gets more). Which one reproduces $g(x_i, 0.25)$ exactly at the nodes, and why?